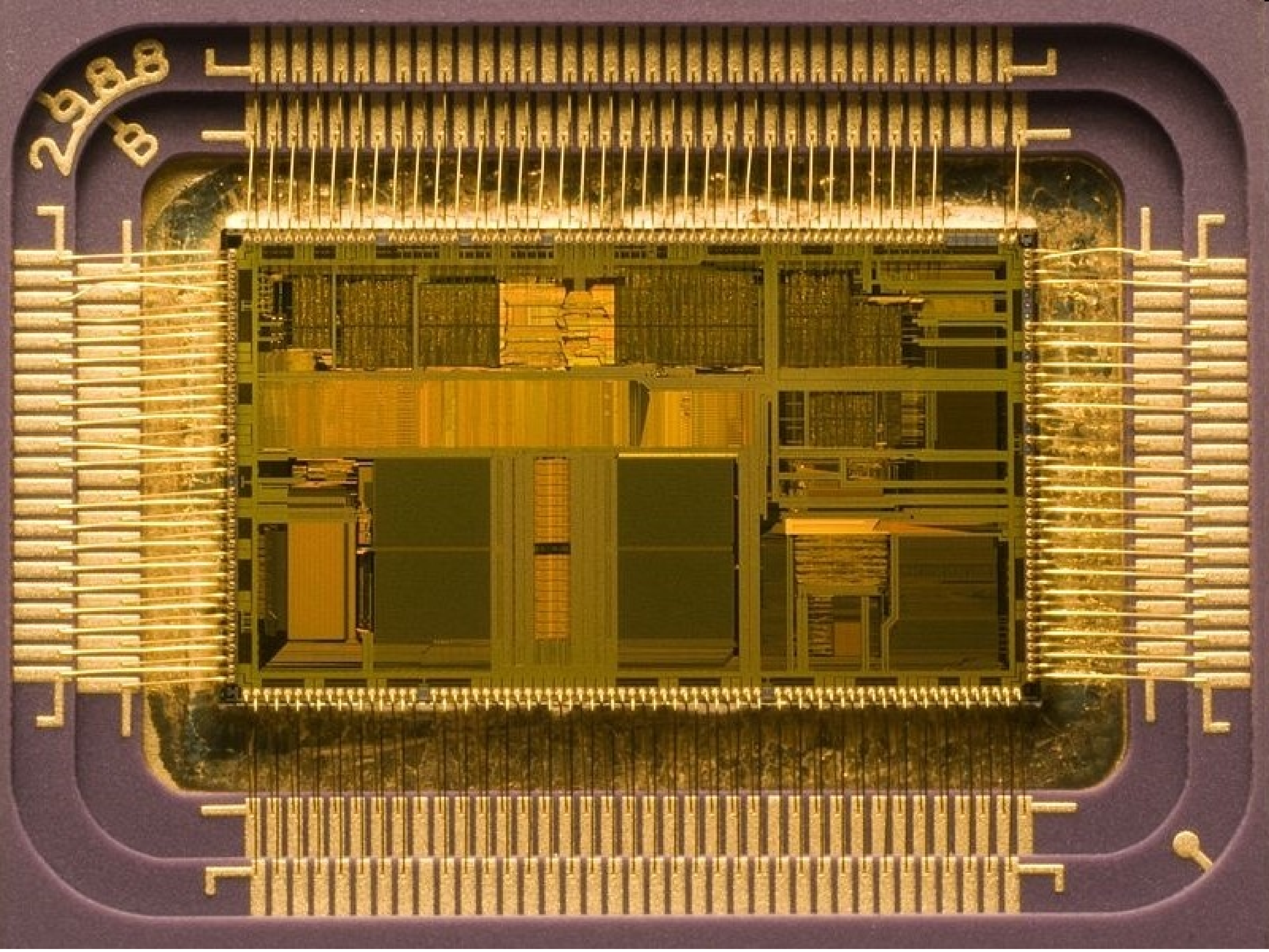
Advanced Micro-Controller Systems

- Introduction, Microcontroller basics, Si technology
- 32-bit Microcontroller technology
- ARM Cortex M Architecture, (CM0 CM3 CM4 CM7)
- Cortex M Microcontroller Instruction Set
- Exception Handling, Interrupts
- Real-time Operating Systems (RTOS)
- Memory Management, protected memory, MPU
- Implementing Calculations, Data Formats, FPU
- I/O, Device drivers
- Application Design
- Case Studies, Advanced Design Patterns

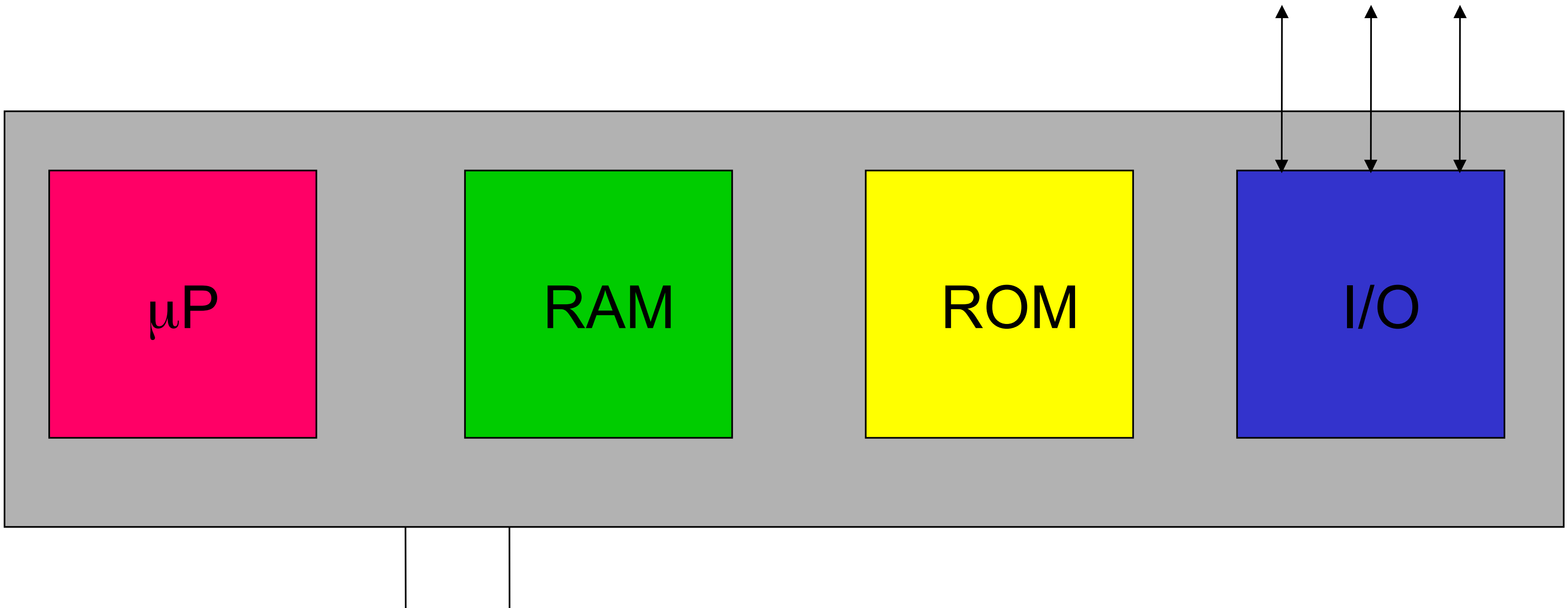
https://skript.eit.h-da.de/ams/

Monocrystalline Silicon Wafer



Mikro-controller

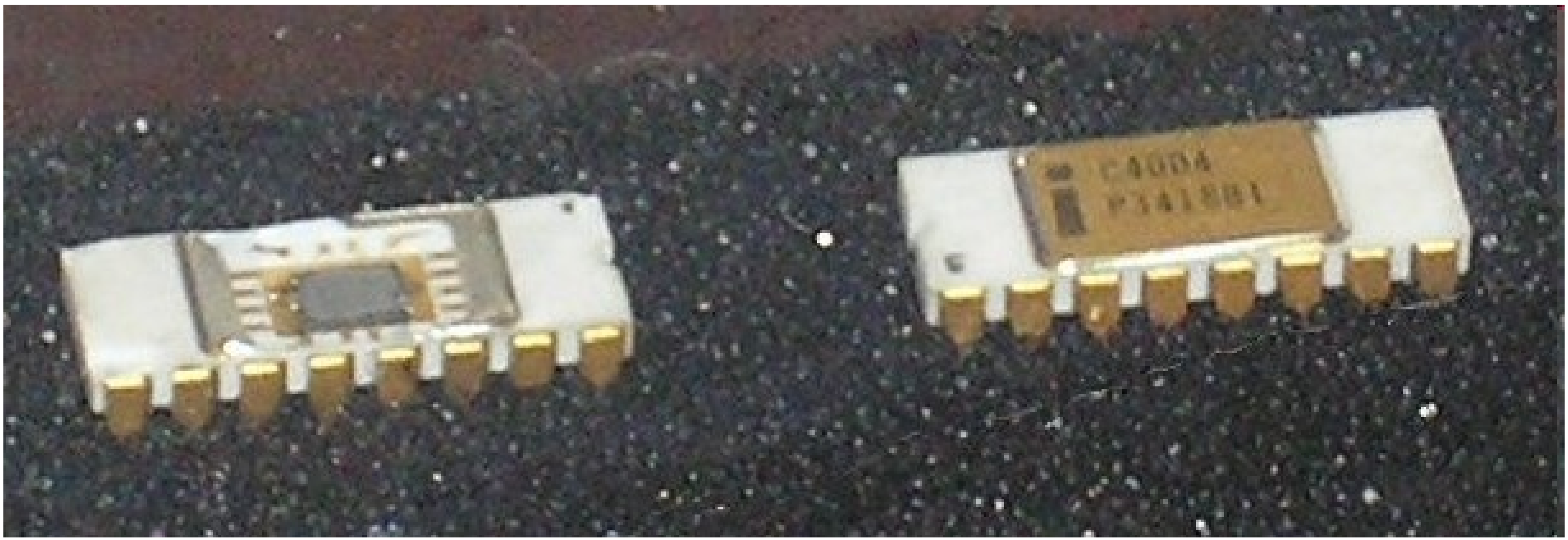
Single Chip Computer System



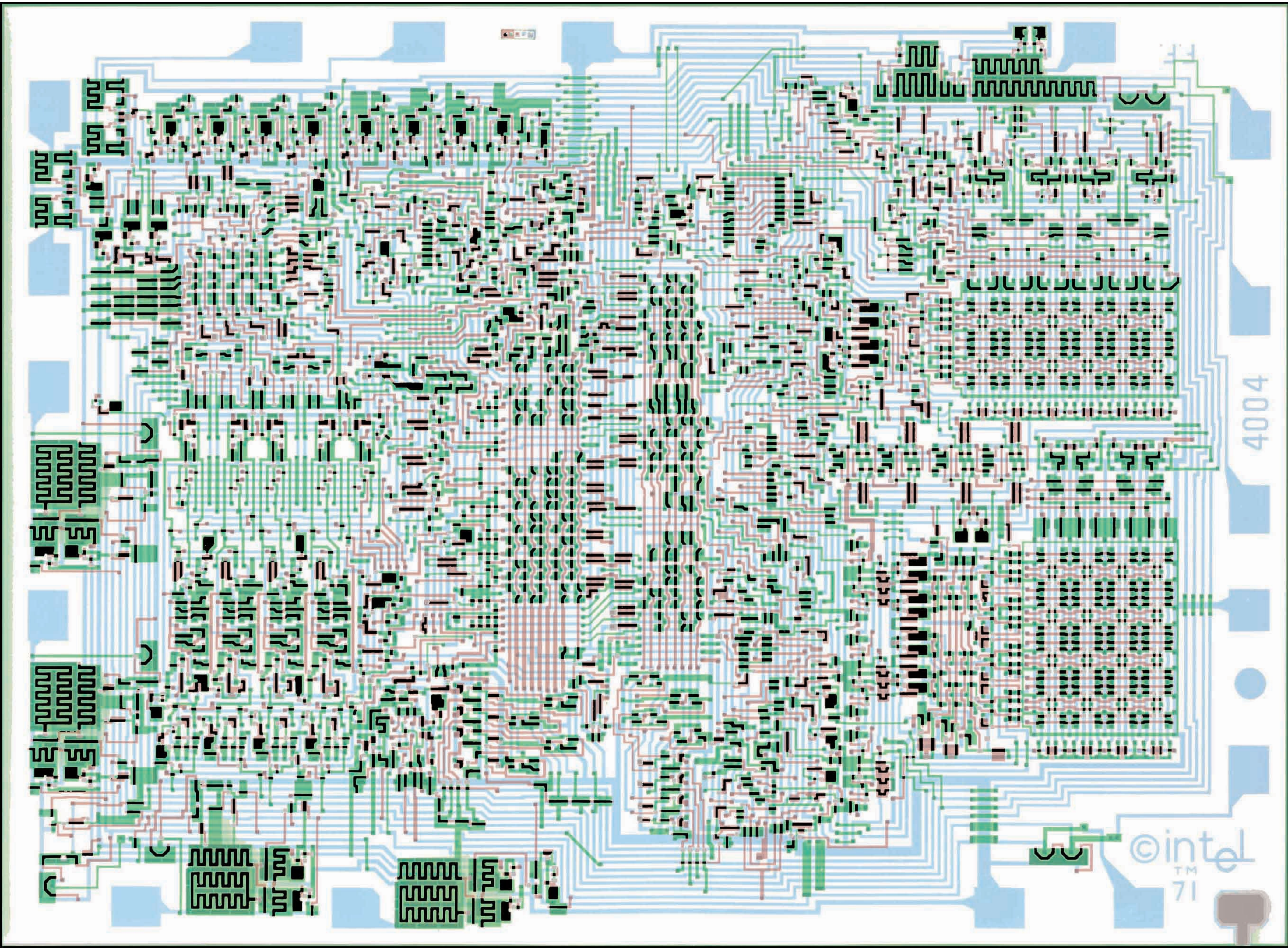
+ -

Battery

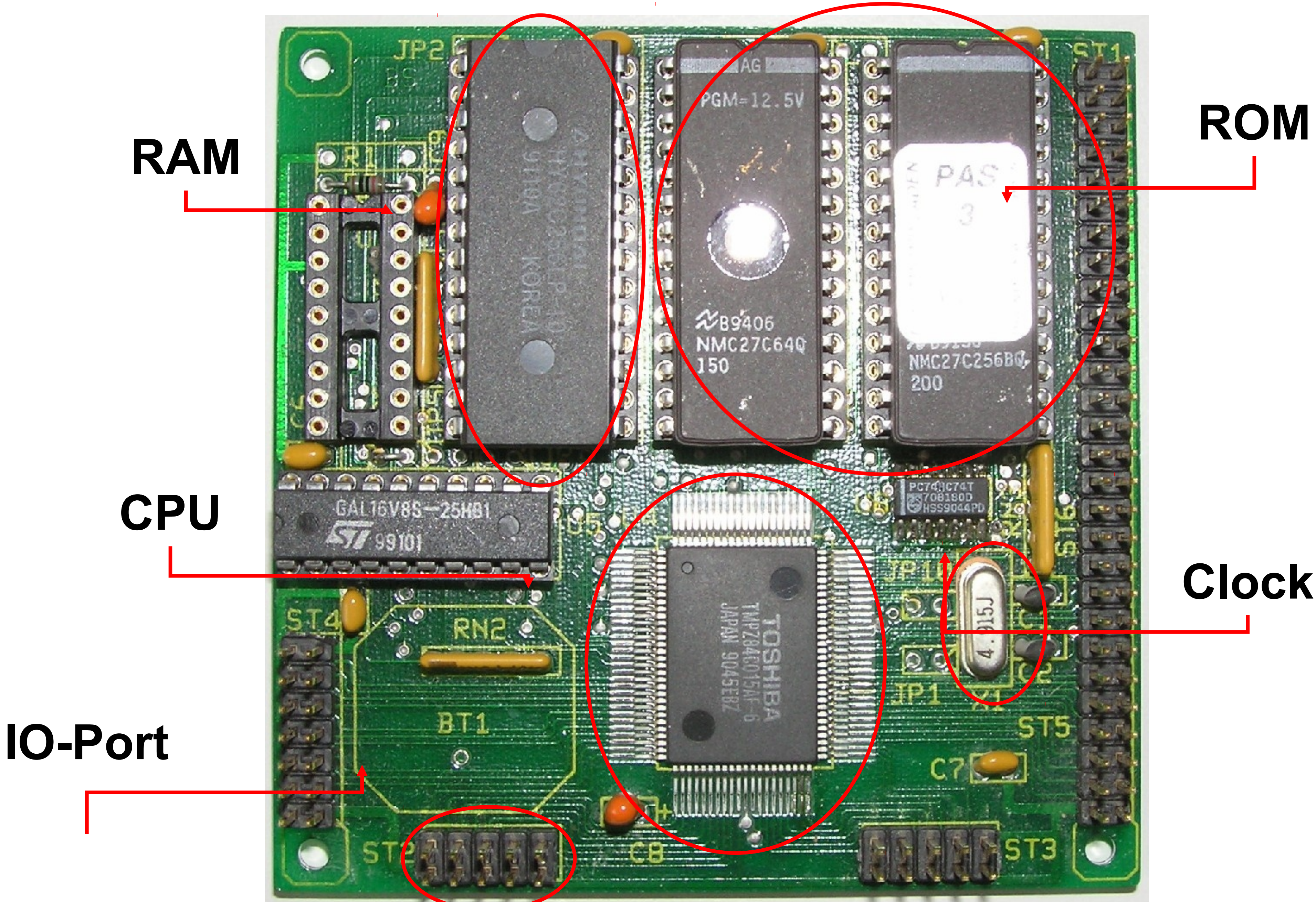
First Microprocessor



First 4 Bit Mikroprocessor: Intel 4004 (1971)

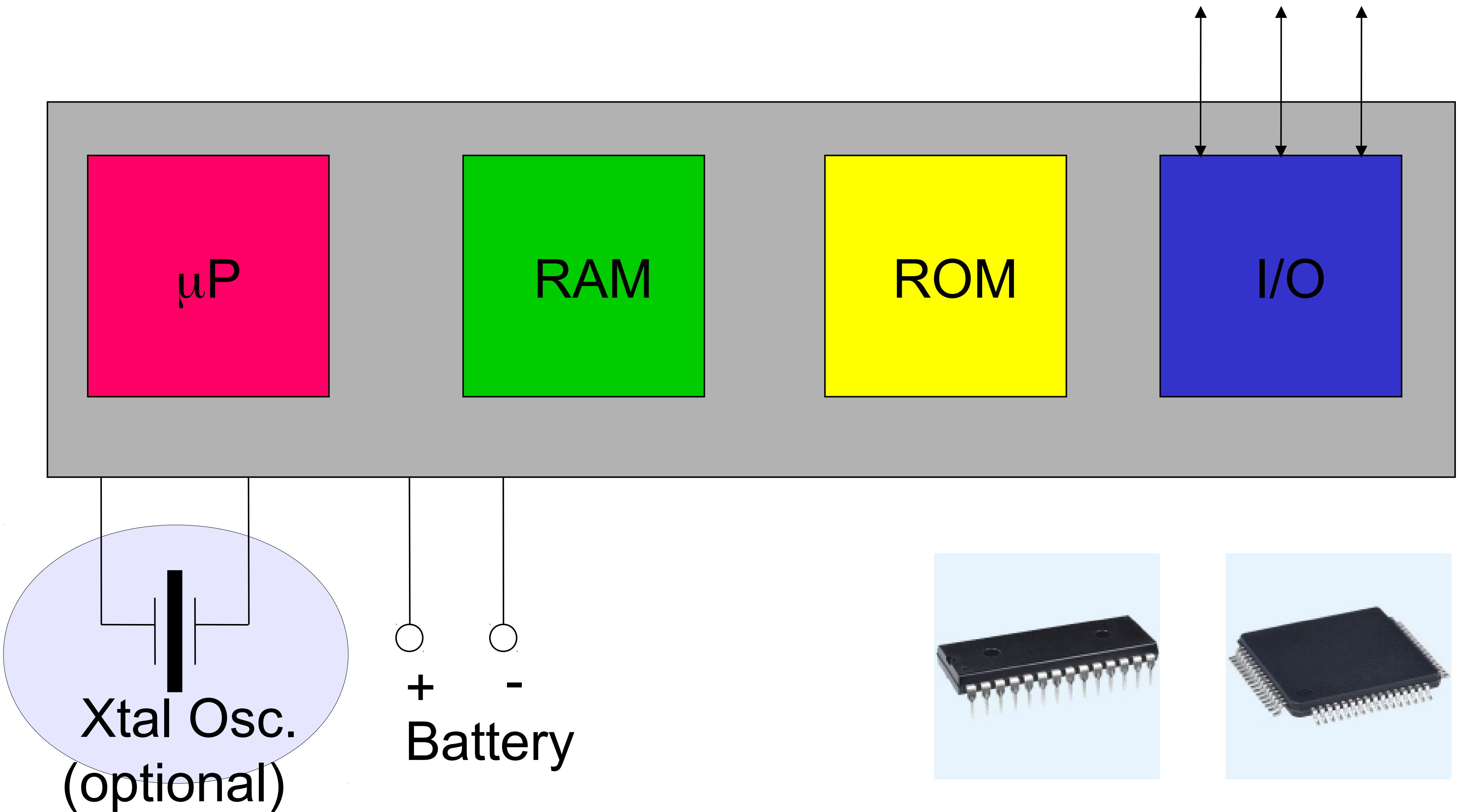

4004
©intel
TM
71

Microprocessor-Board (Zilog Z80)

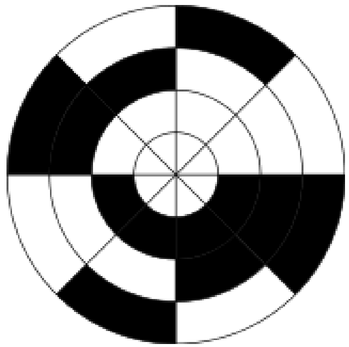


Mikrocontroller

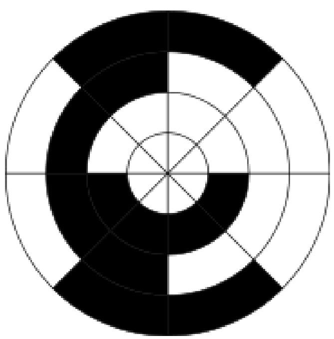
Single Chip Computer System



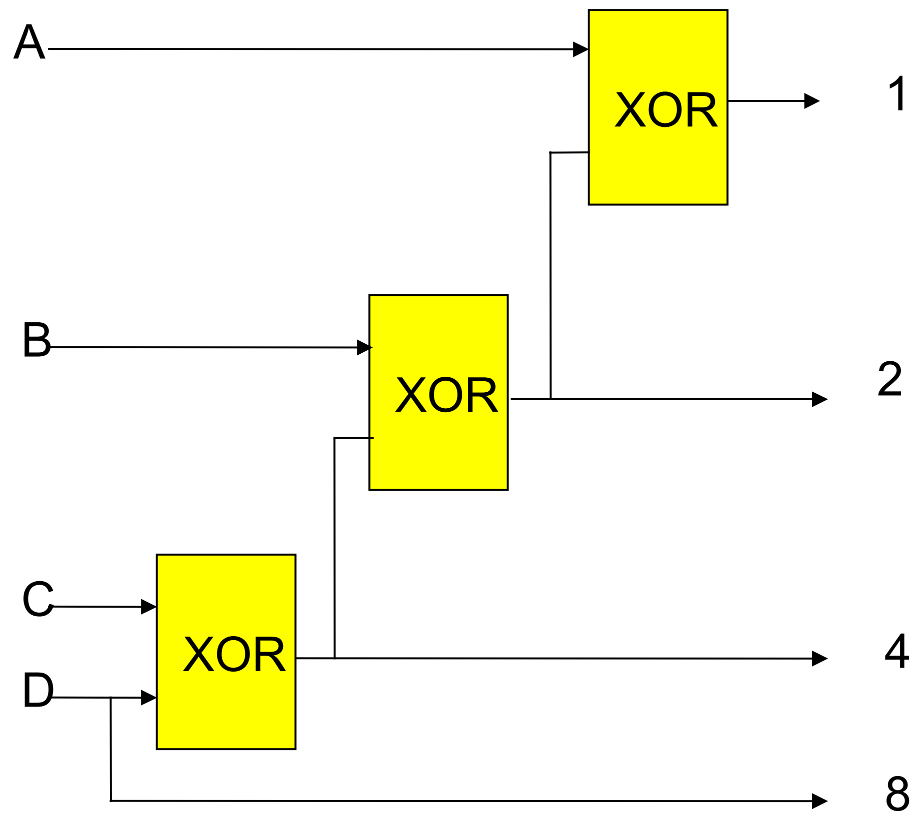
Why do we need a clock ?



Binary code

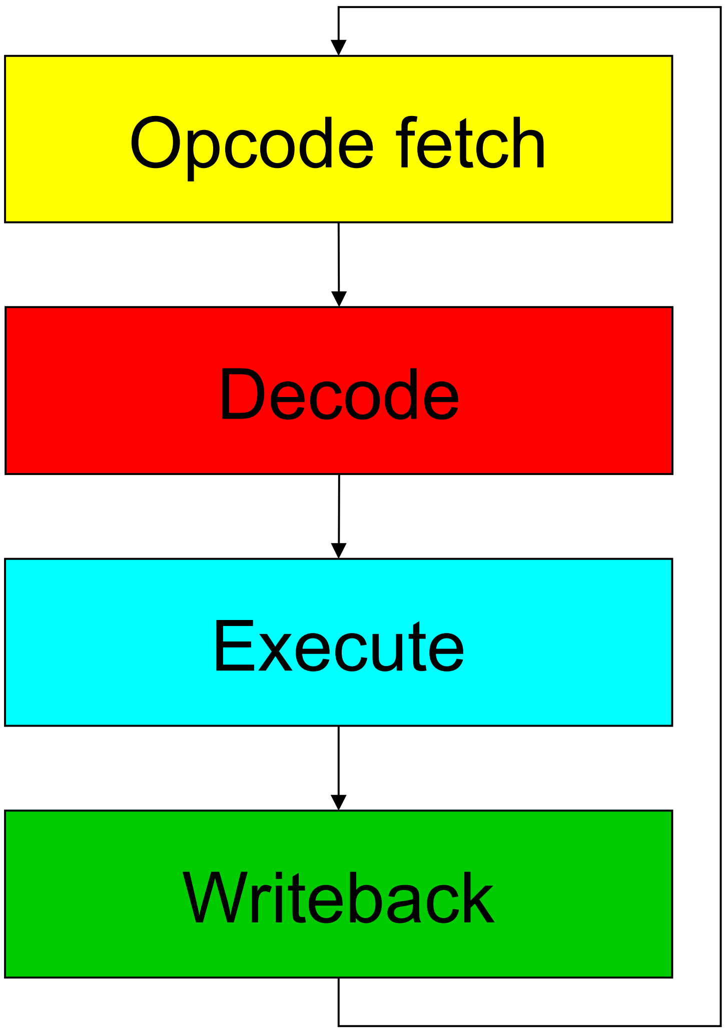


Graycode



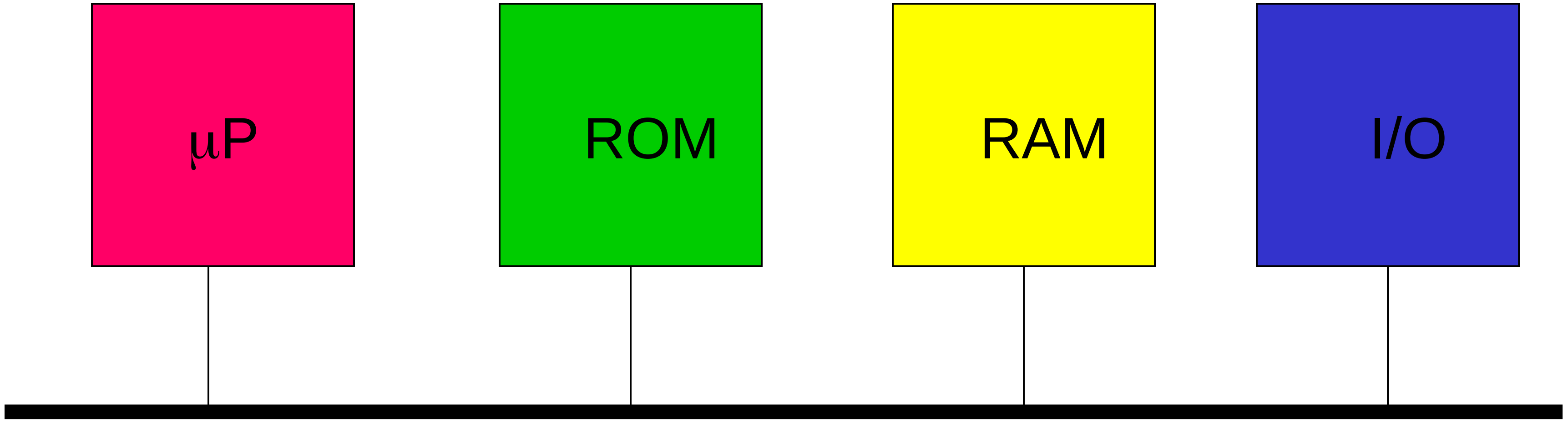
Single Machine Cycle

(Classic RISC pipeline)

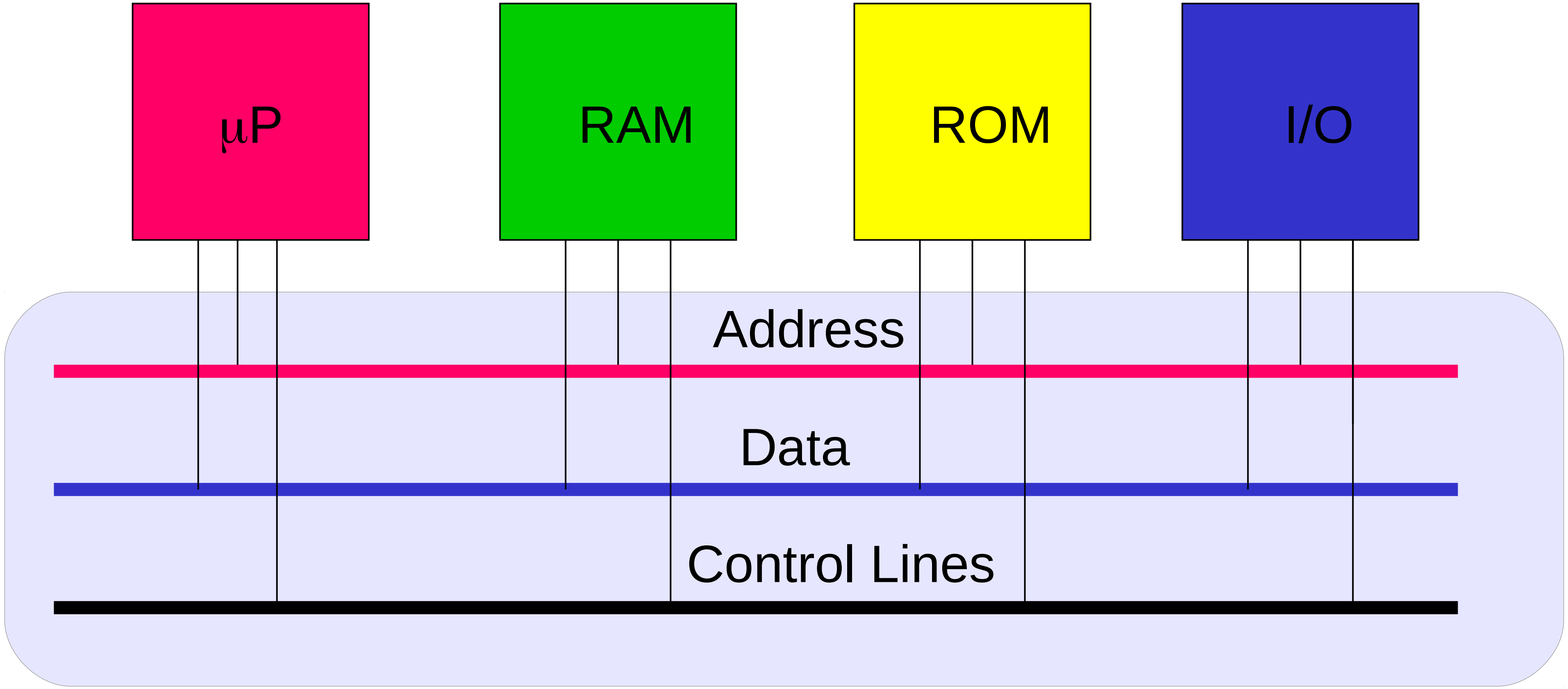


Microprocessor, Microcomputer

Von Neumann Architecture

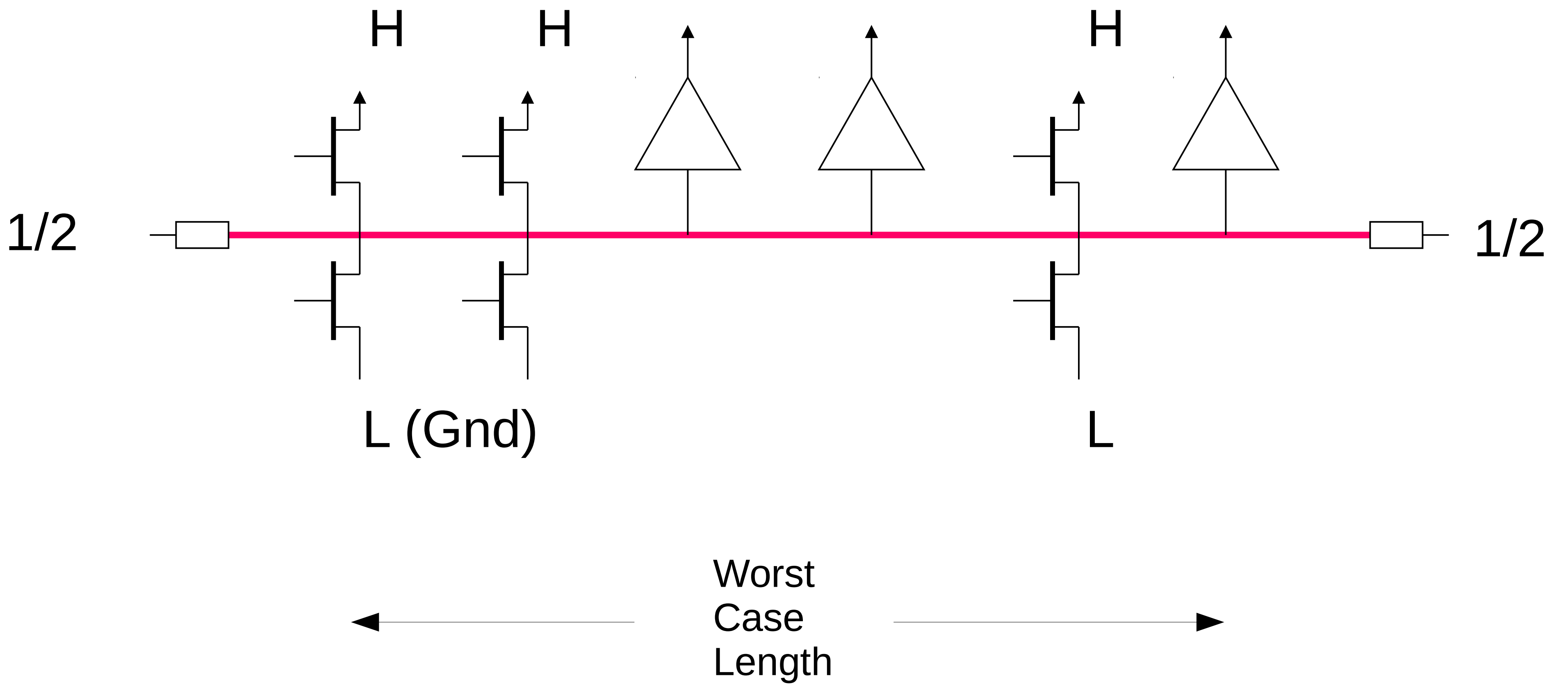


Data and Adress Bus



All three lines are normally multi-directional !

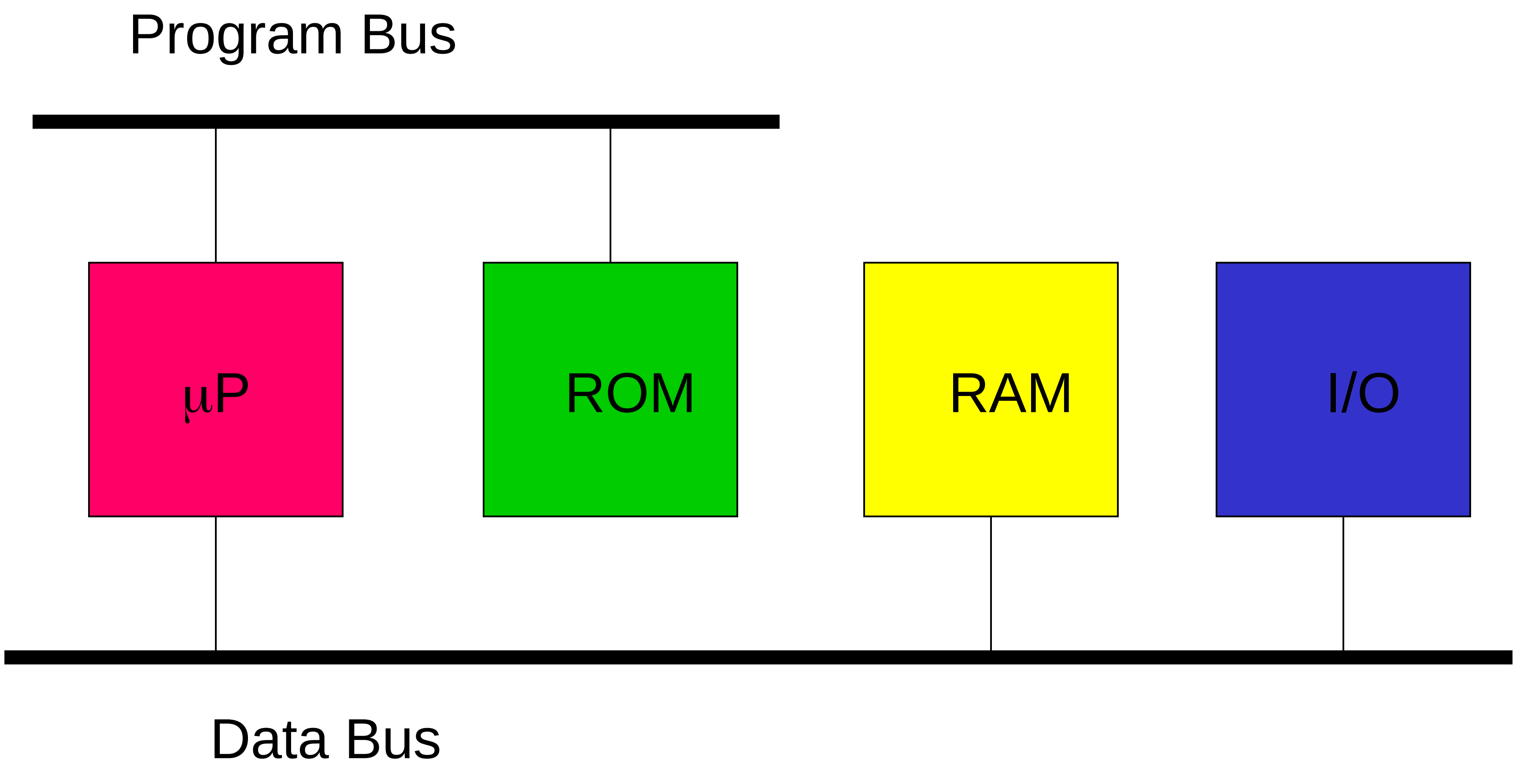
Single Bus Line



Remember: c = 300000 km/s

Microprocessor

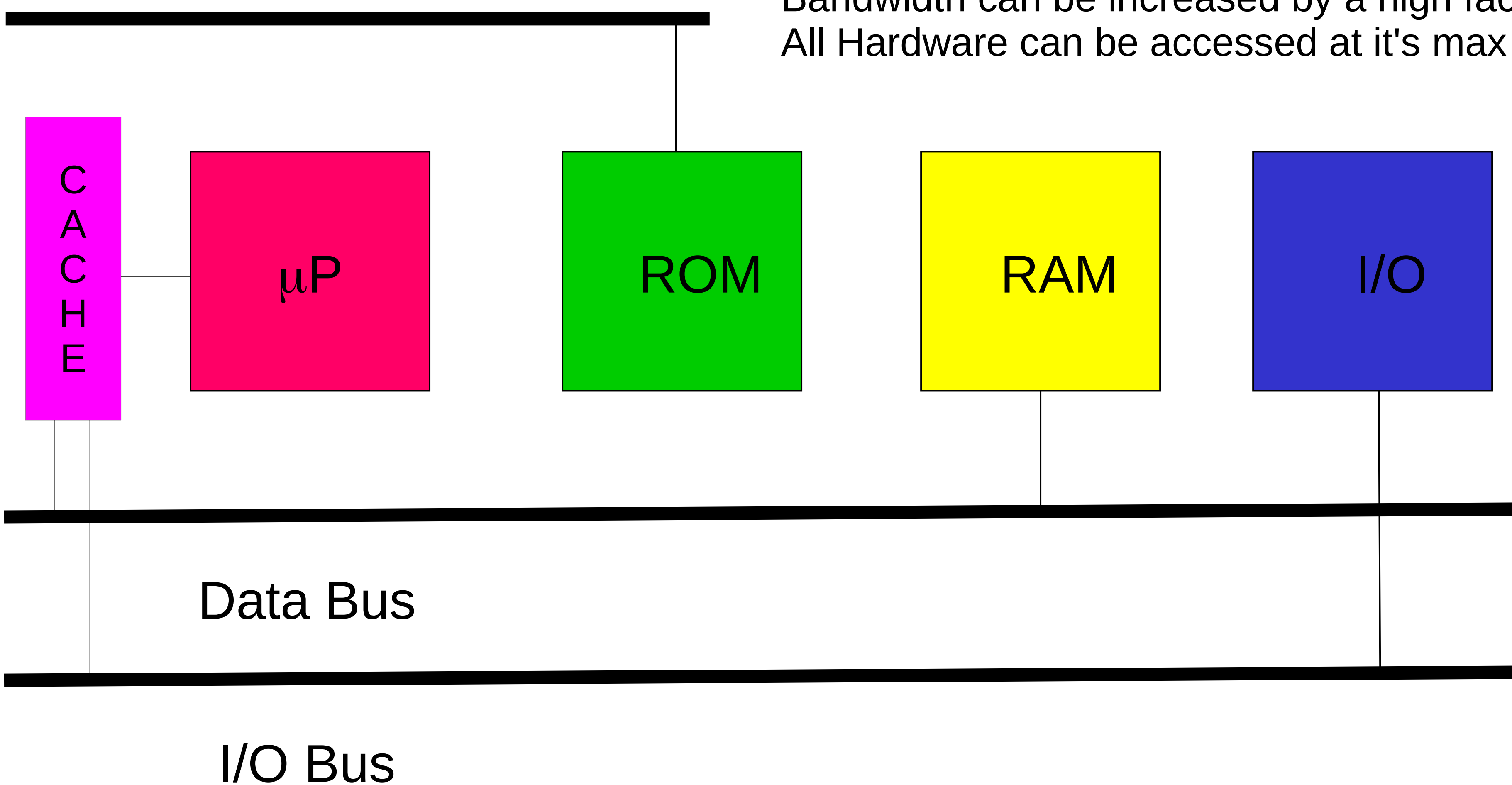
Harvard Architecture



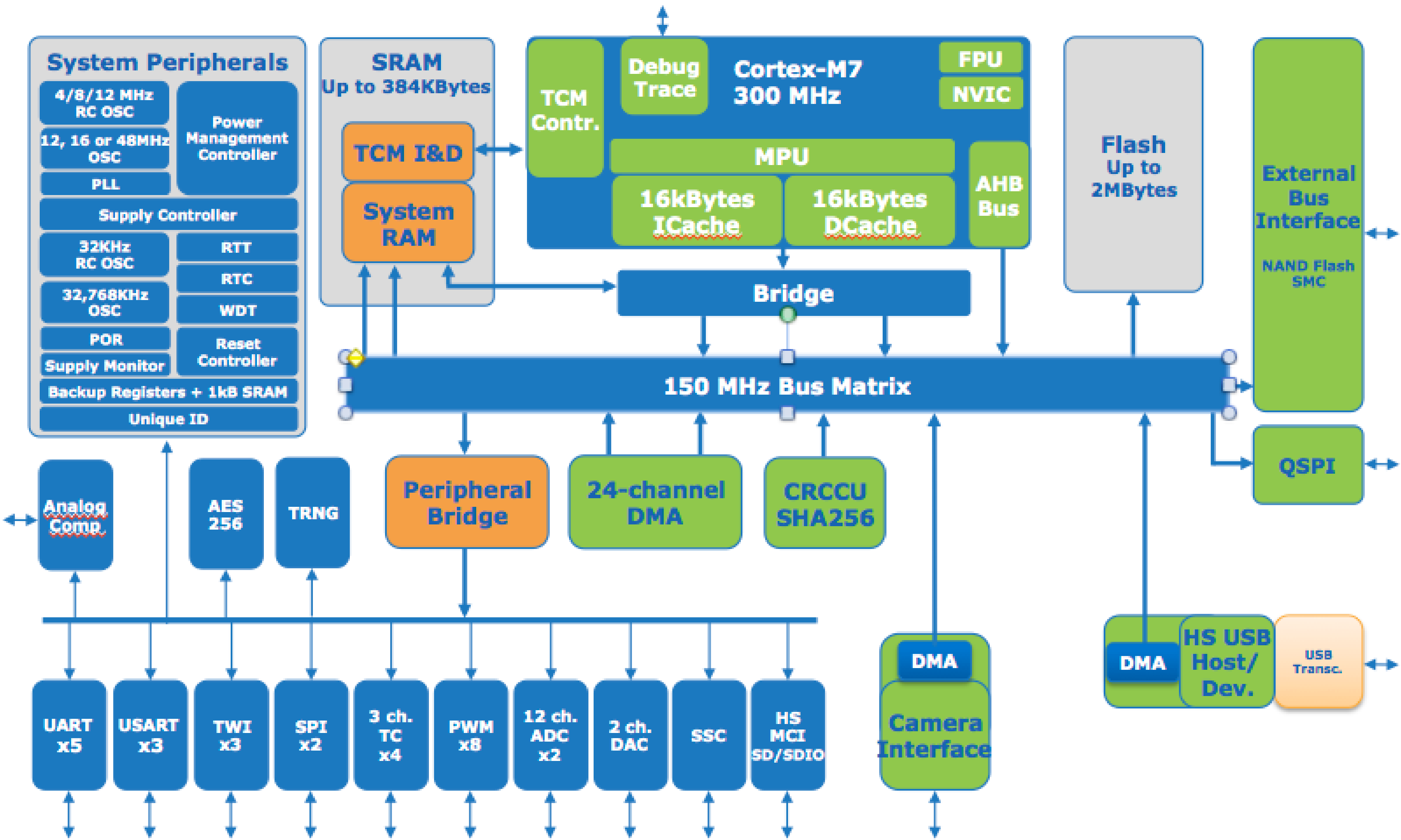
Microprocessor

Modified Harvard Architecture

Advantages:
Single address space can be achieved
Bandwidth can be increased by a high factor
All Hardware can be accessed at it's max speed



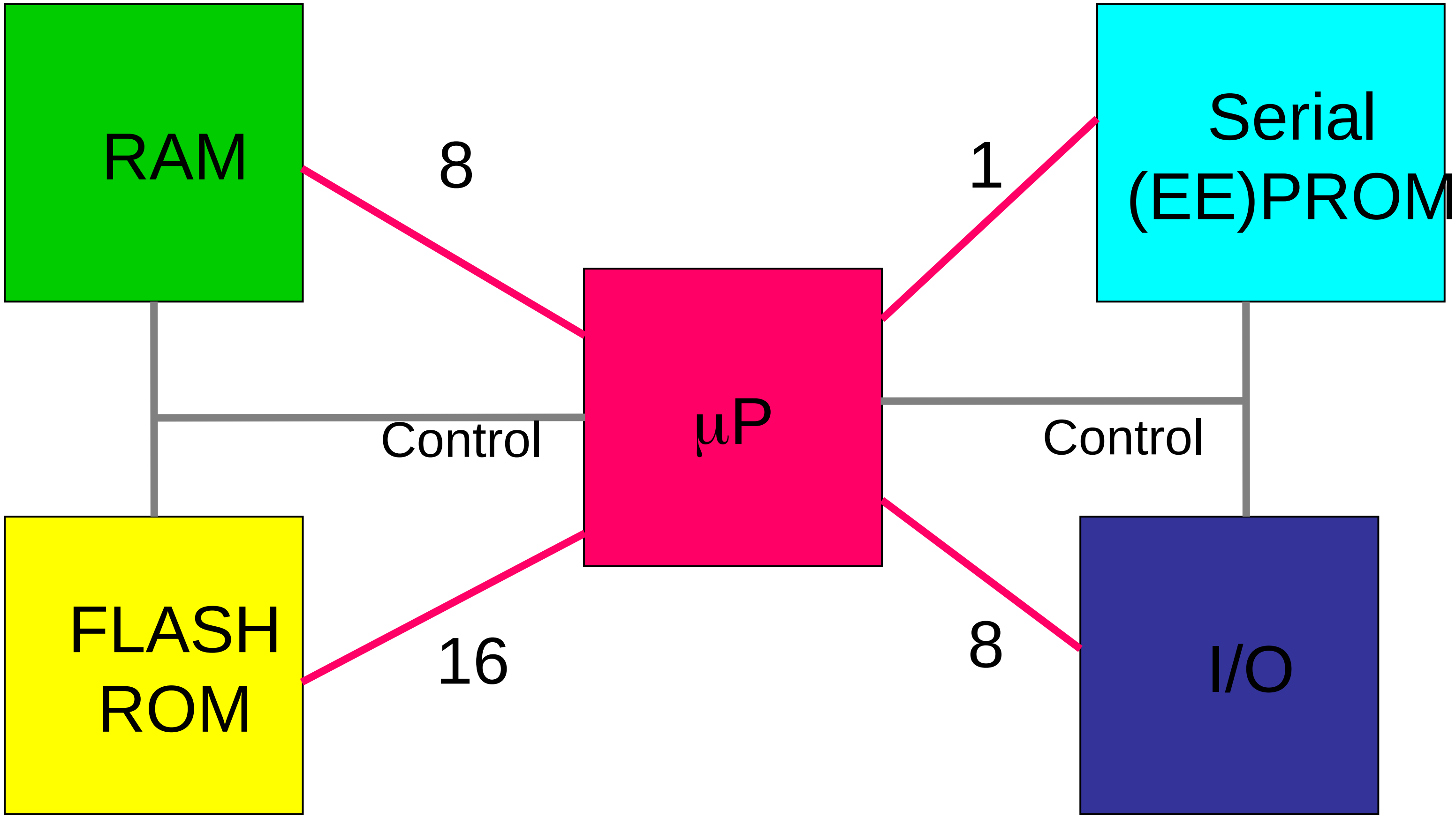
SAM S70



Atmel.com

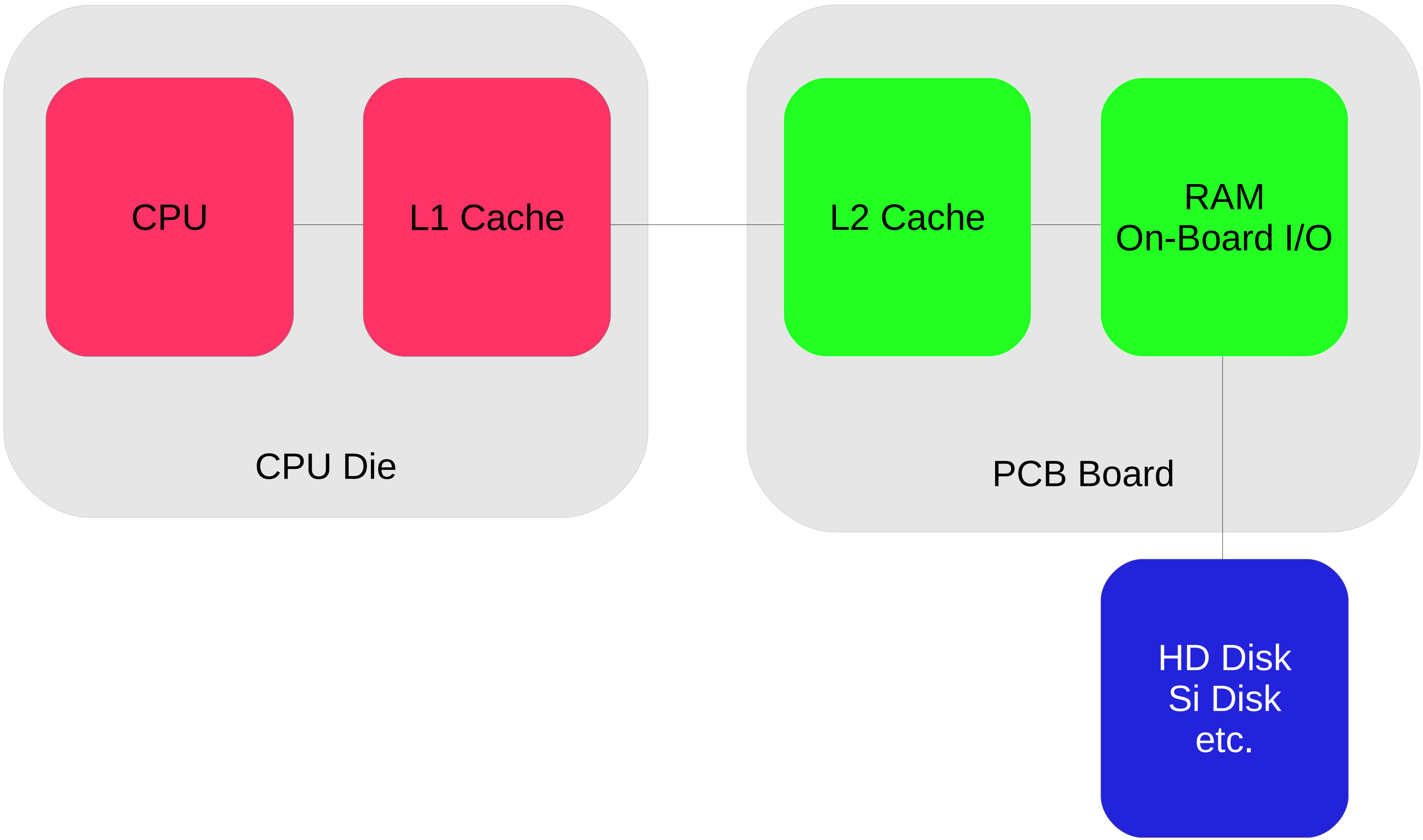
Bus Width

AVR 8-bit Microcontroller

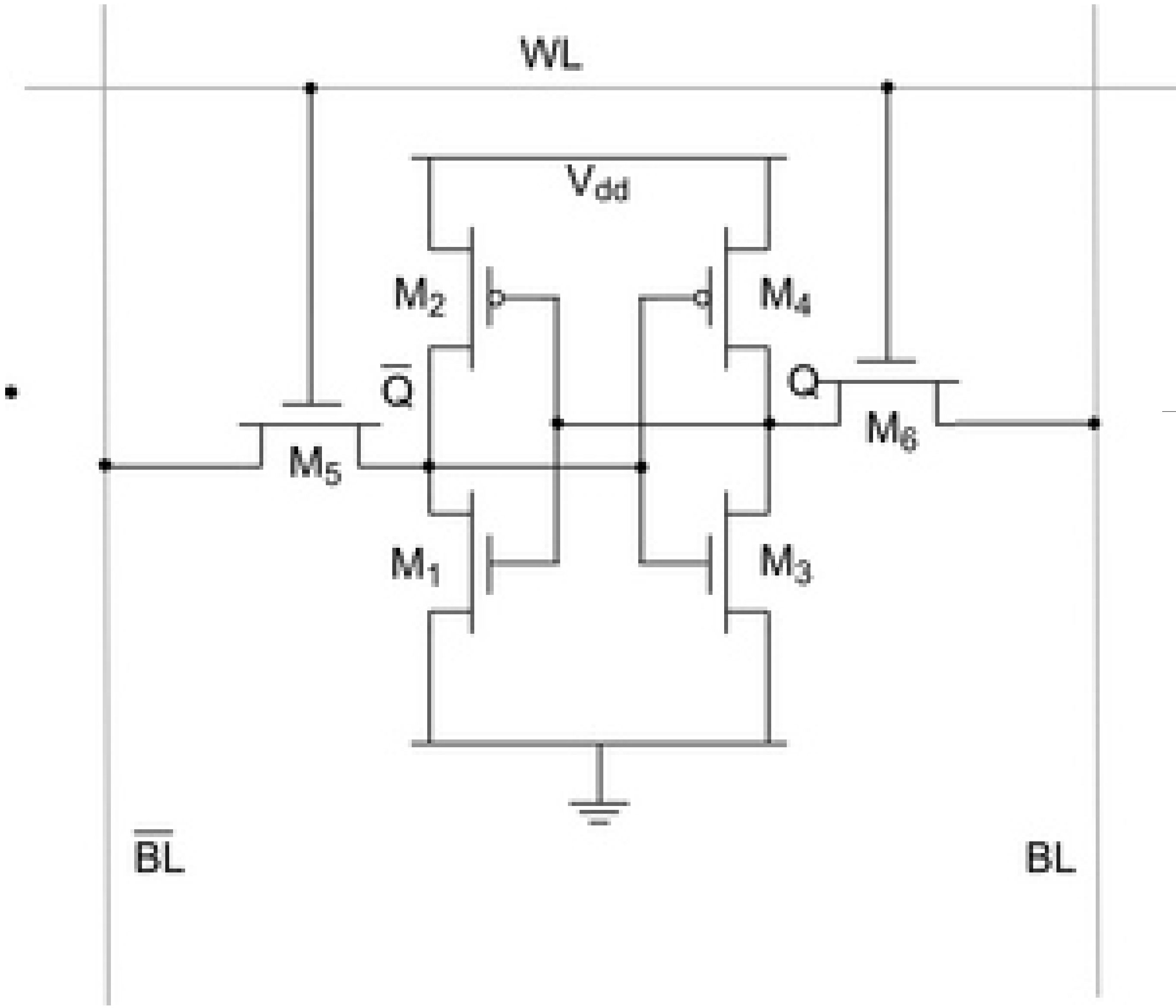


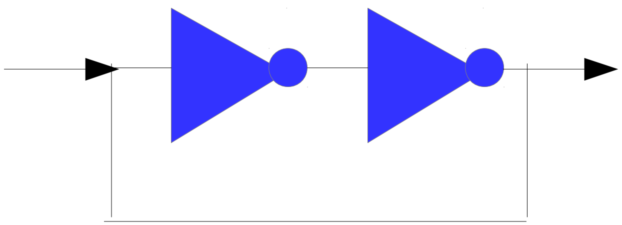
Remark: This is a quite peculiar design ...

PC / Server Memory



Static SRAM Cell



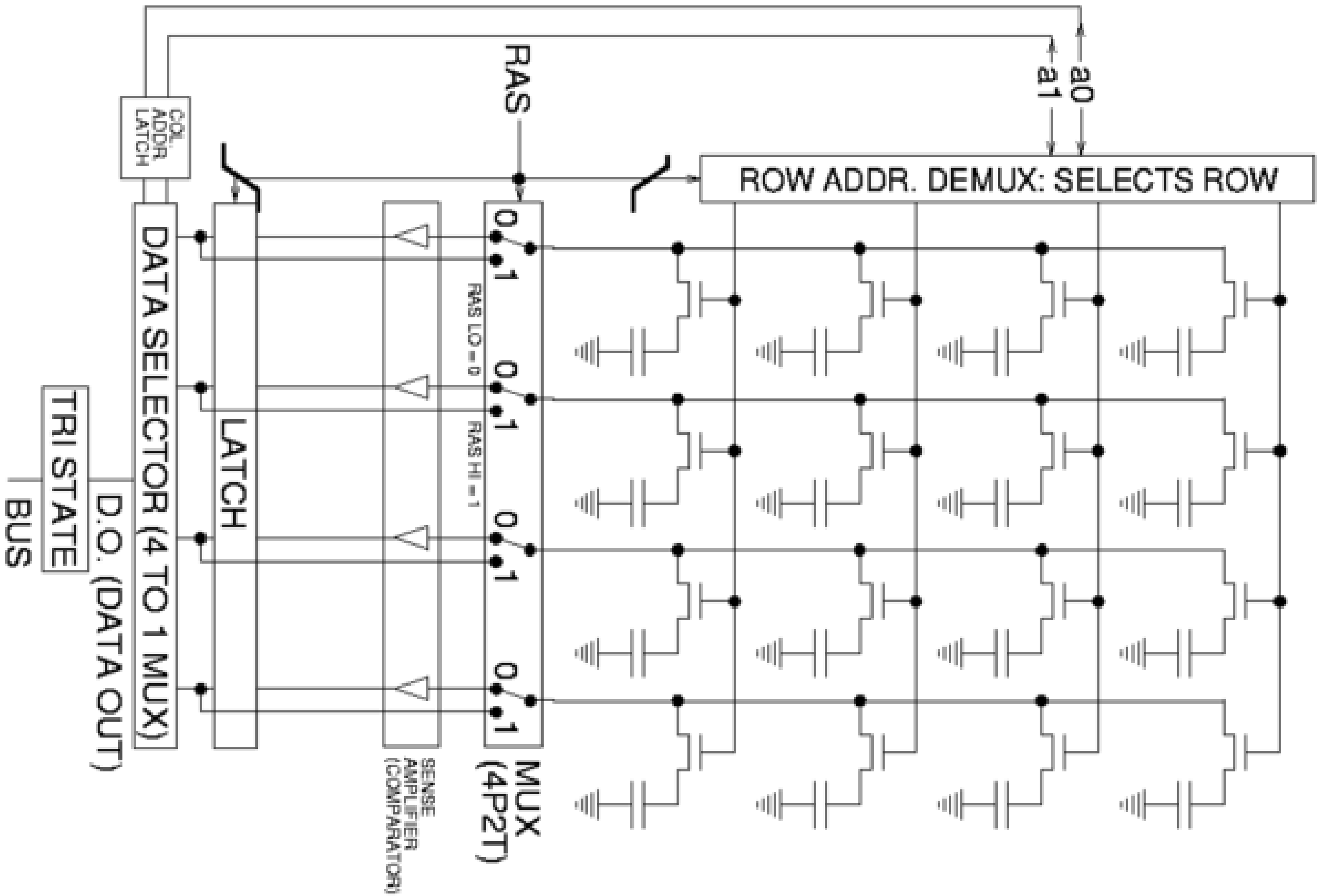


Feedback-loop made from 2 inverters

Needs (very little) power to retain memory content

Source: Wikipedia

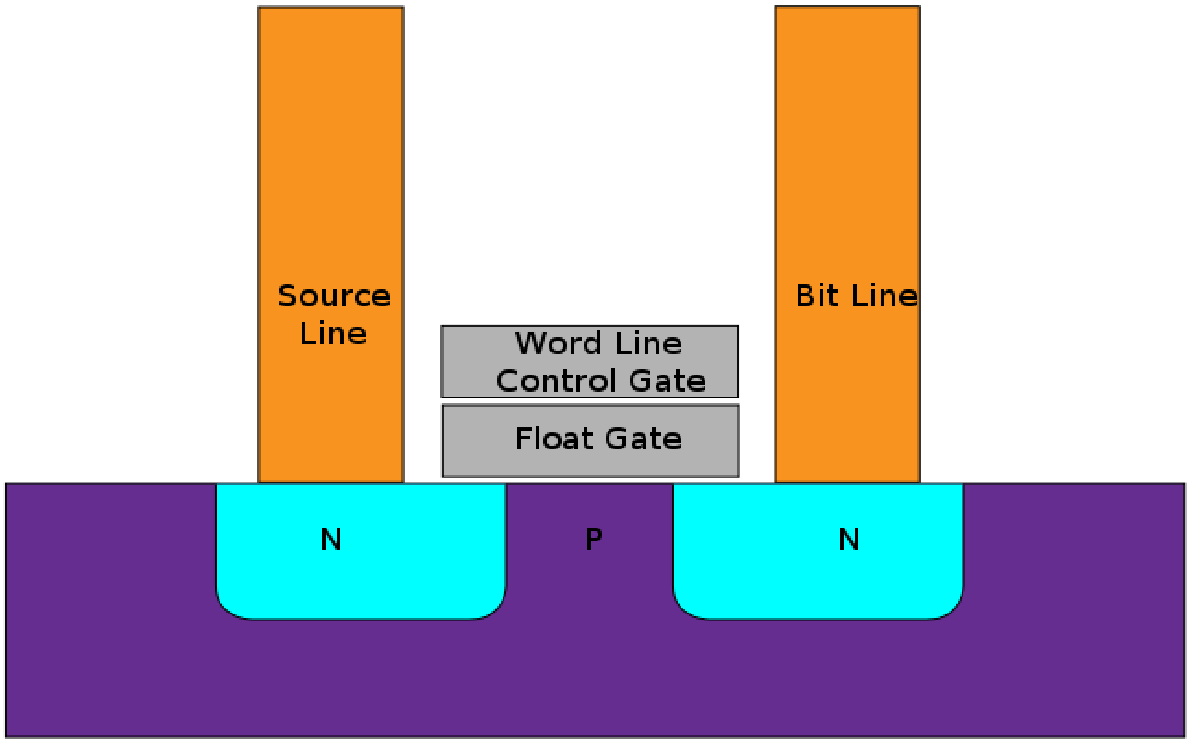
Dynamic RAM



Dynamic RAM needs refresh resulting in high power consumption
Eventually good for big computers, useless for embedded systems.

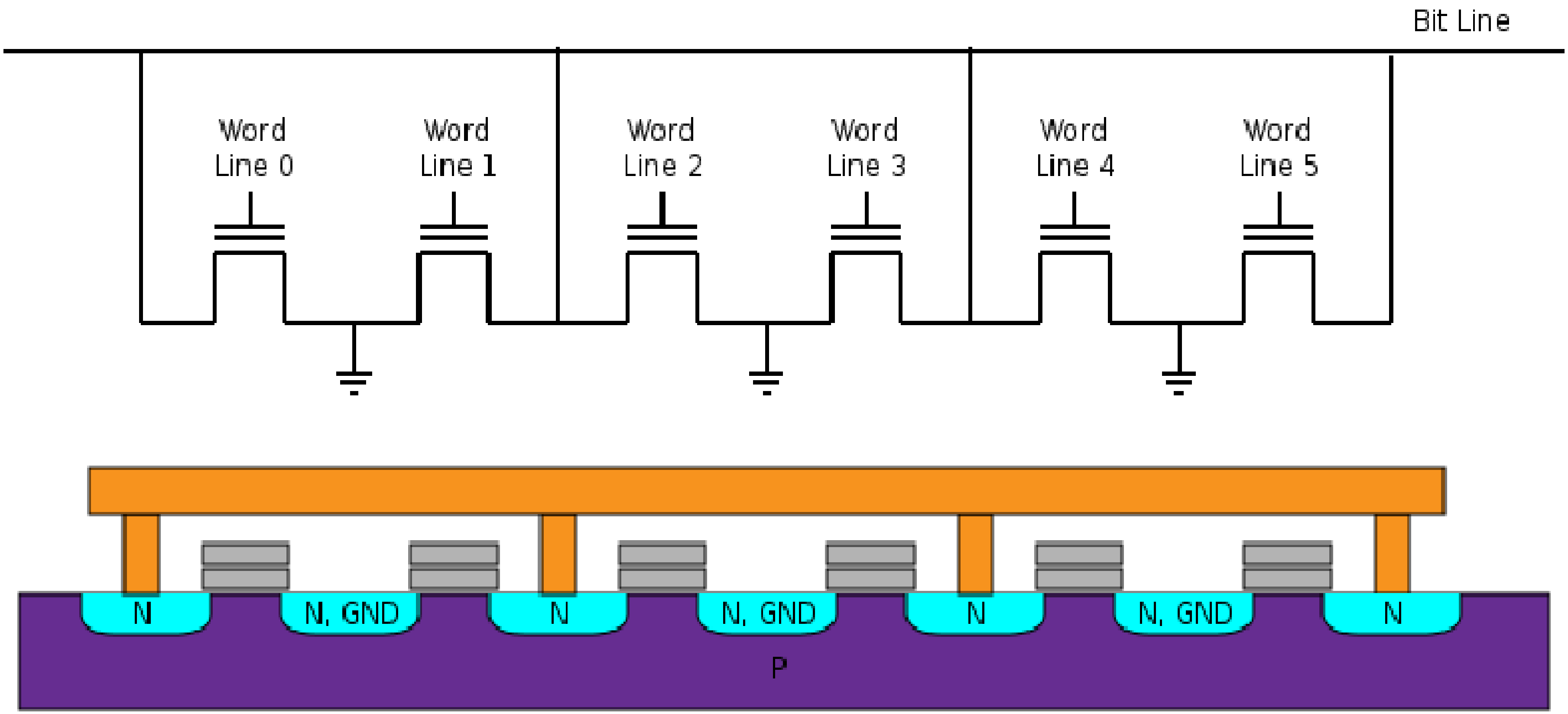
Source: Wikipedia

FLASH Memory



Source: Wikipedia

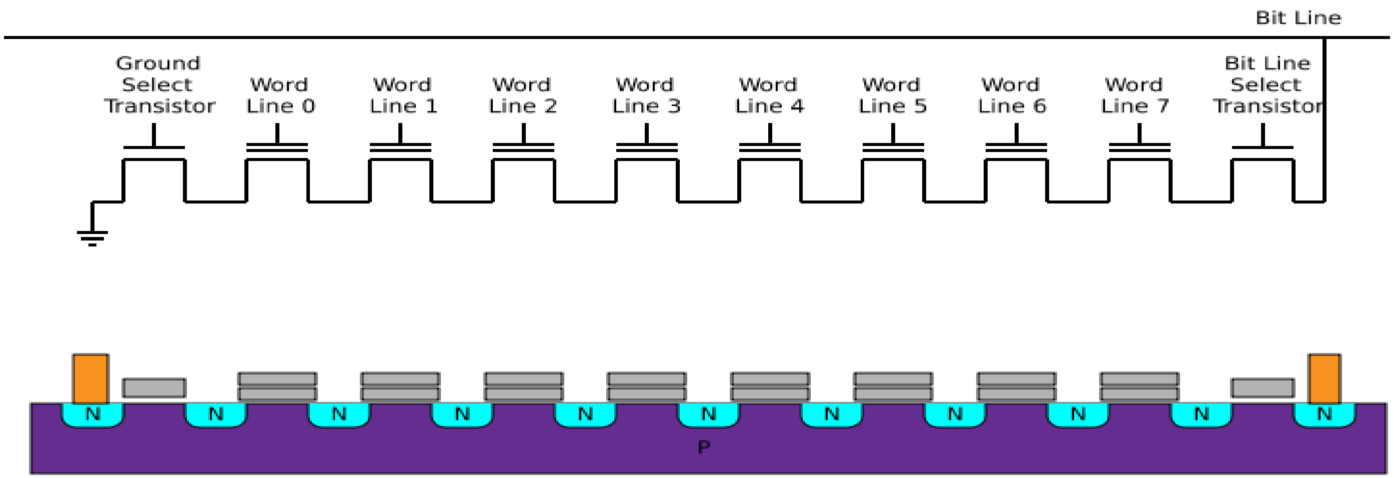
FLASH Memory
NOR FLASH



Used for program memory in modern controllers

Source: Wikipedia

FLASH Memory
NAND Flash



Read and written in big blocks (a.e. 256 bytes) at a time

Used in silicon disks

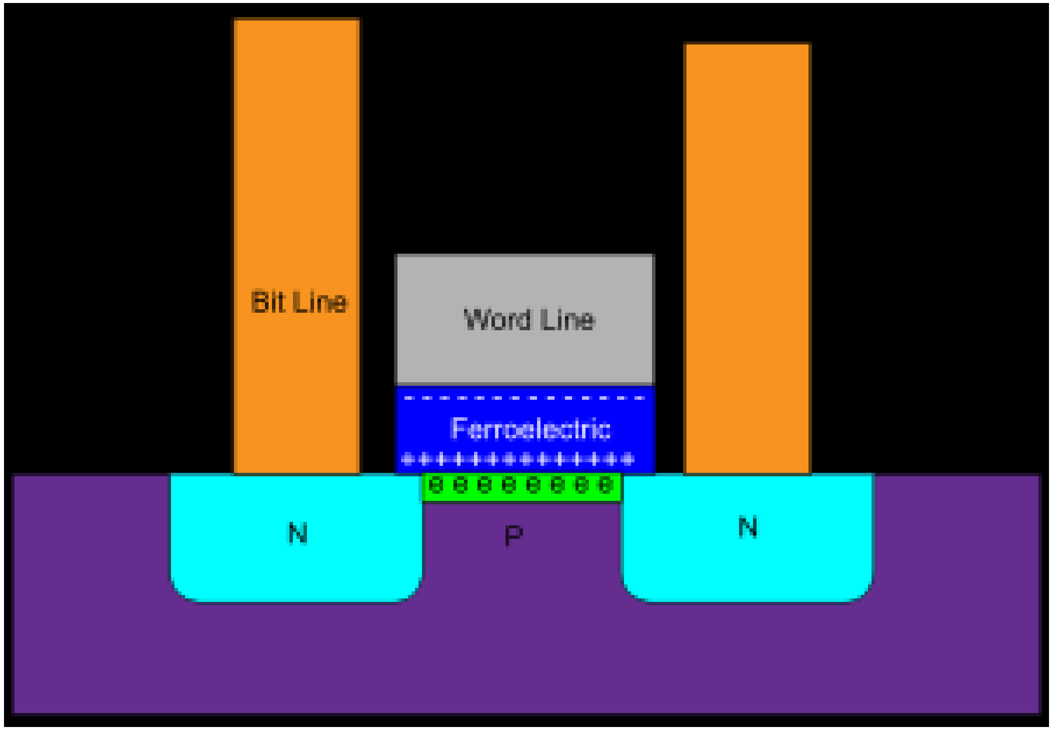
Not for micro-controller program memory

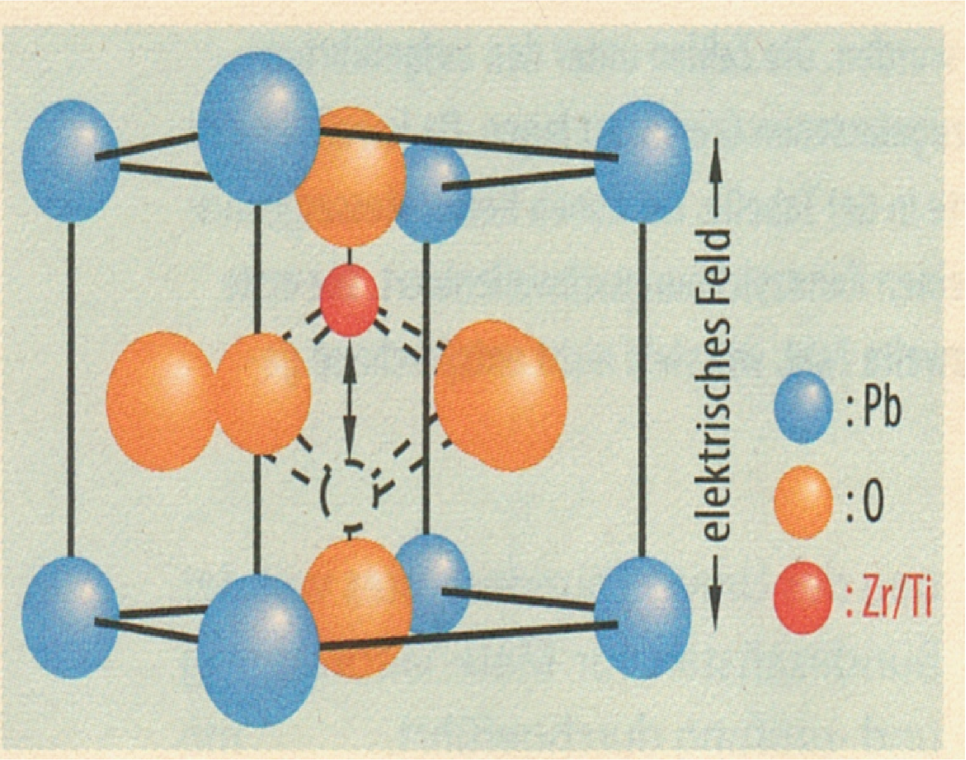
Source: Wikipedia

Source: Wikipedia.org

FeRAM
Ferro-electric RAM Cell

Making use of a bistable polarization state



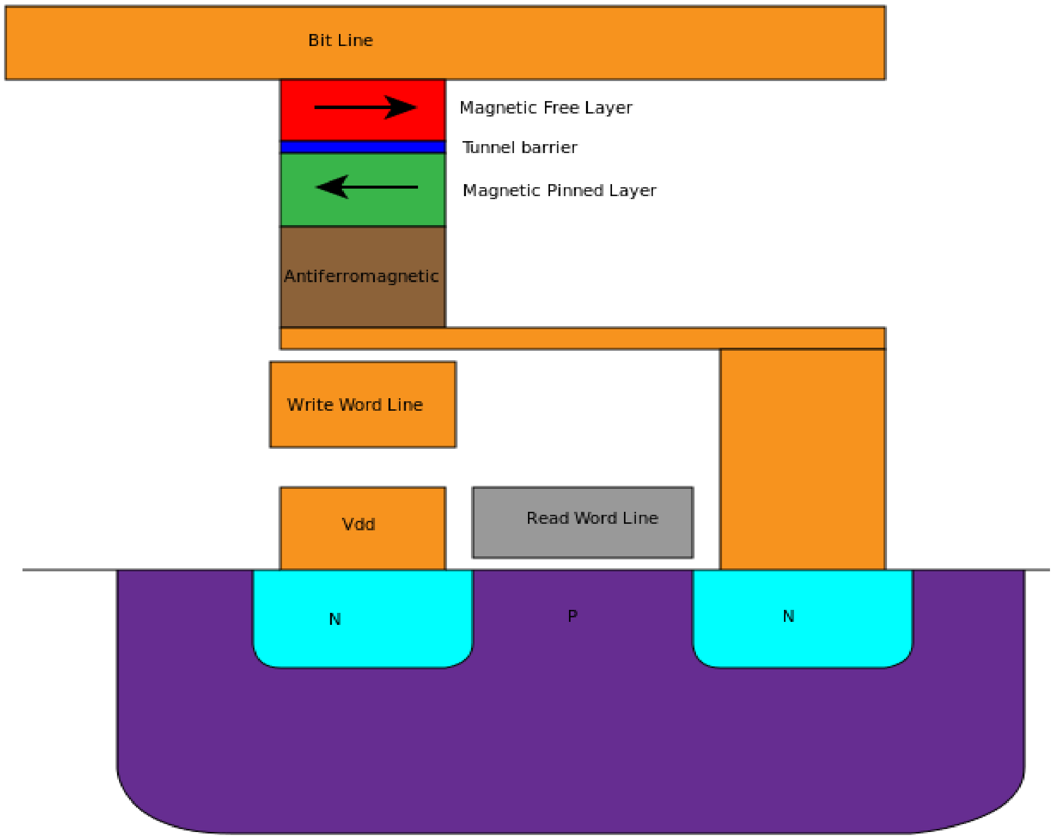


- Nonvolatile
- Fast
- Low writing energy compared to FLASH

Physical:
PZT = Lead zirconium titanate

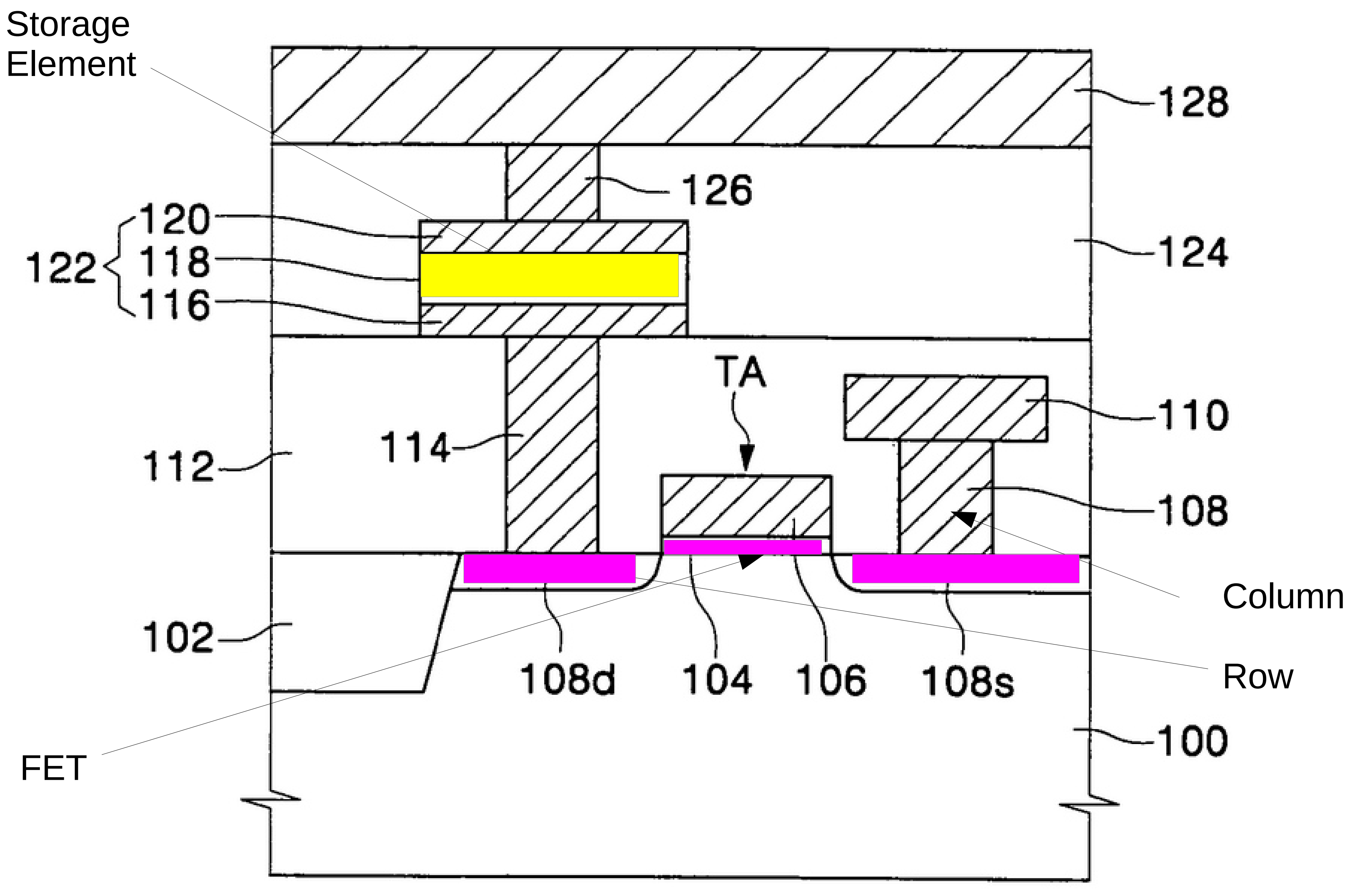
- Pyroelectric
- Piezoelectric
- Ferroelectric

MRAM
Magnetoresistive RAM



Source: Wikipedia

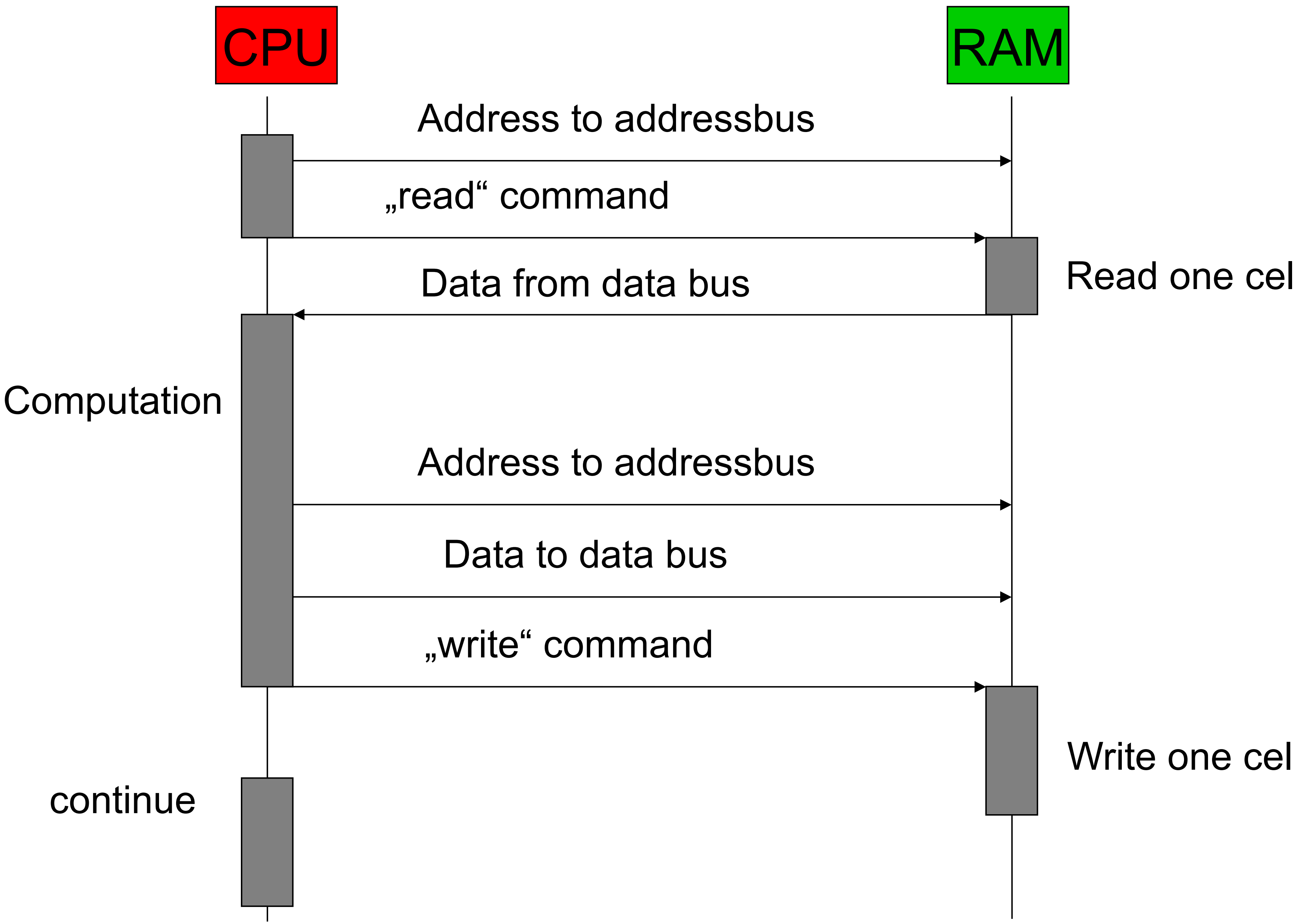
ReRAM
Resistive RAM



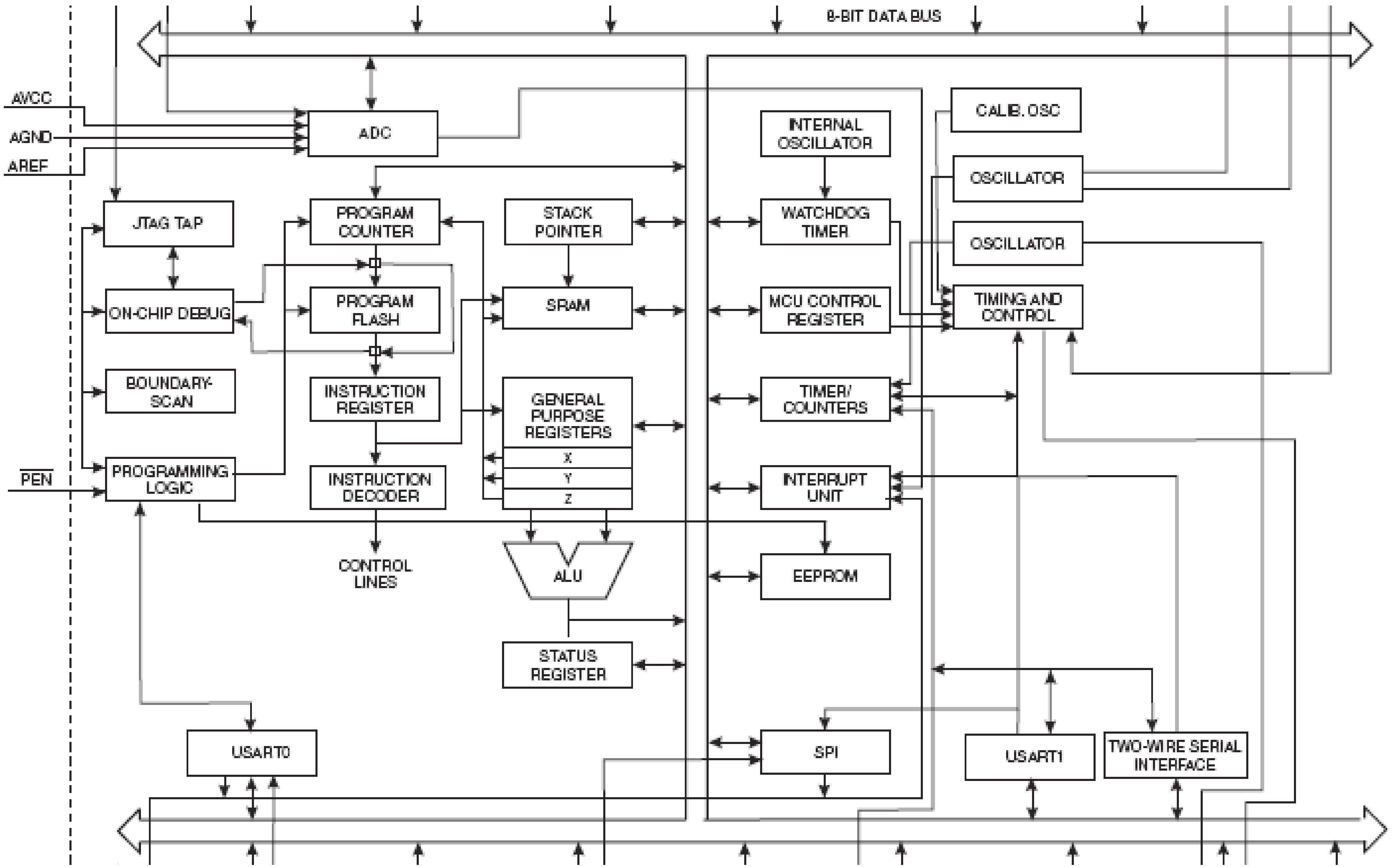
Source: US Patent US7292469B2

Memory Access Cycle

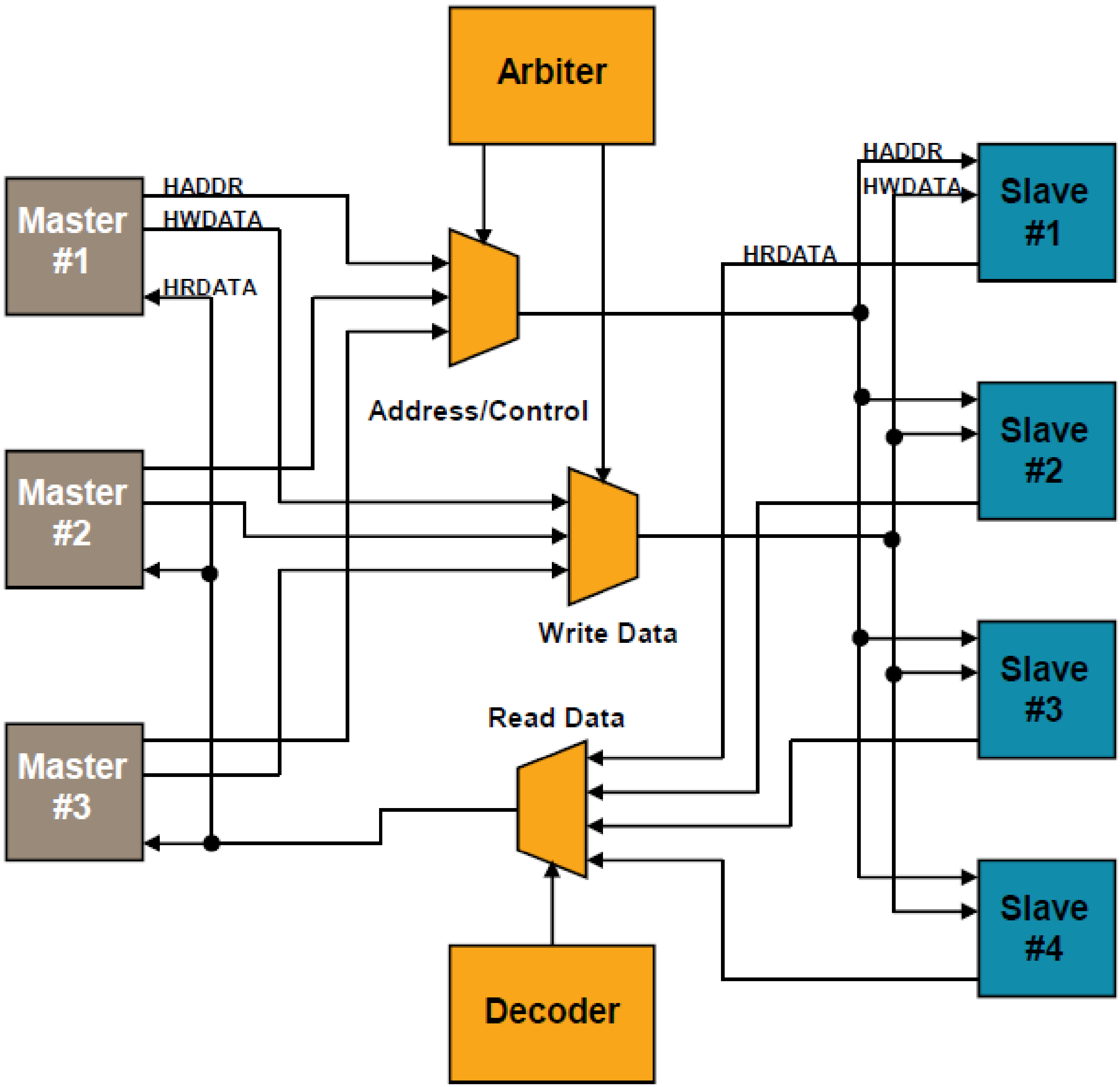
(read - modify - write)



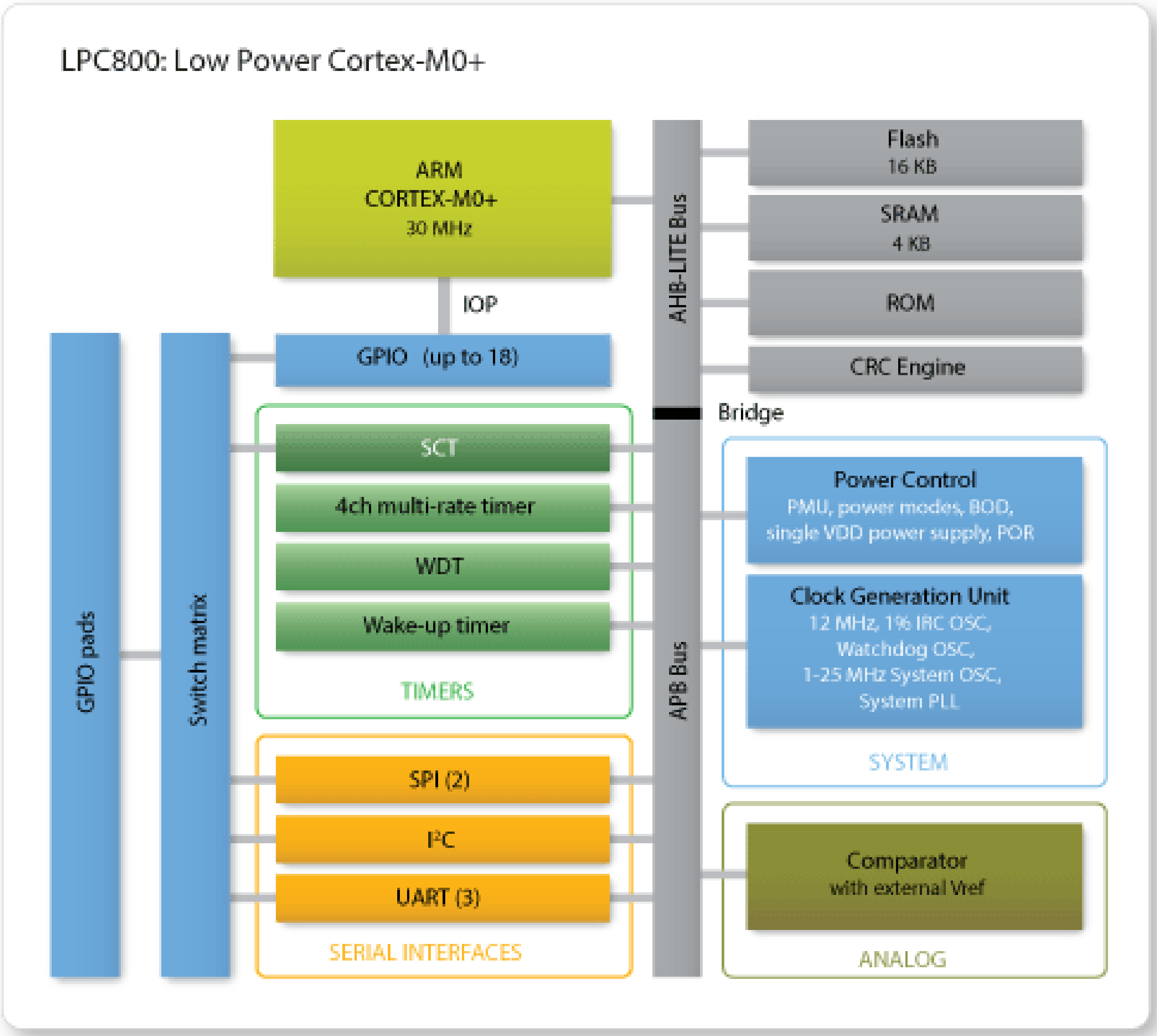
8-bit Microcontroller (ATMEL AVR)



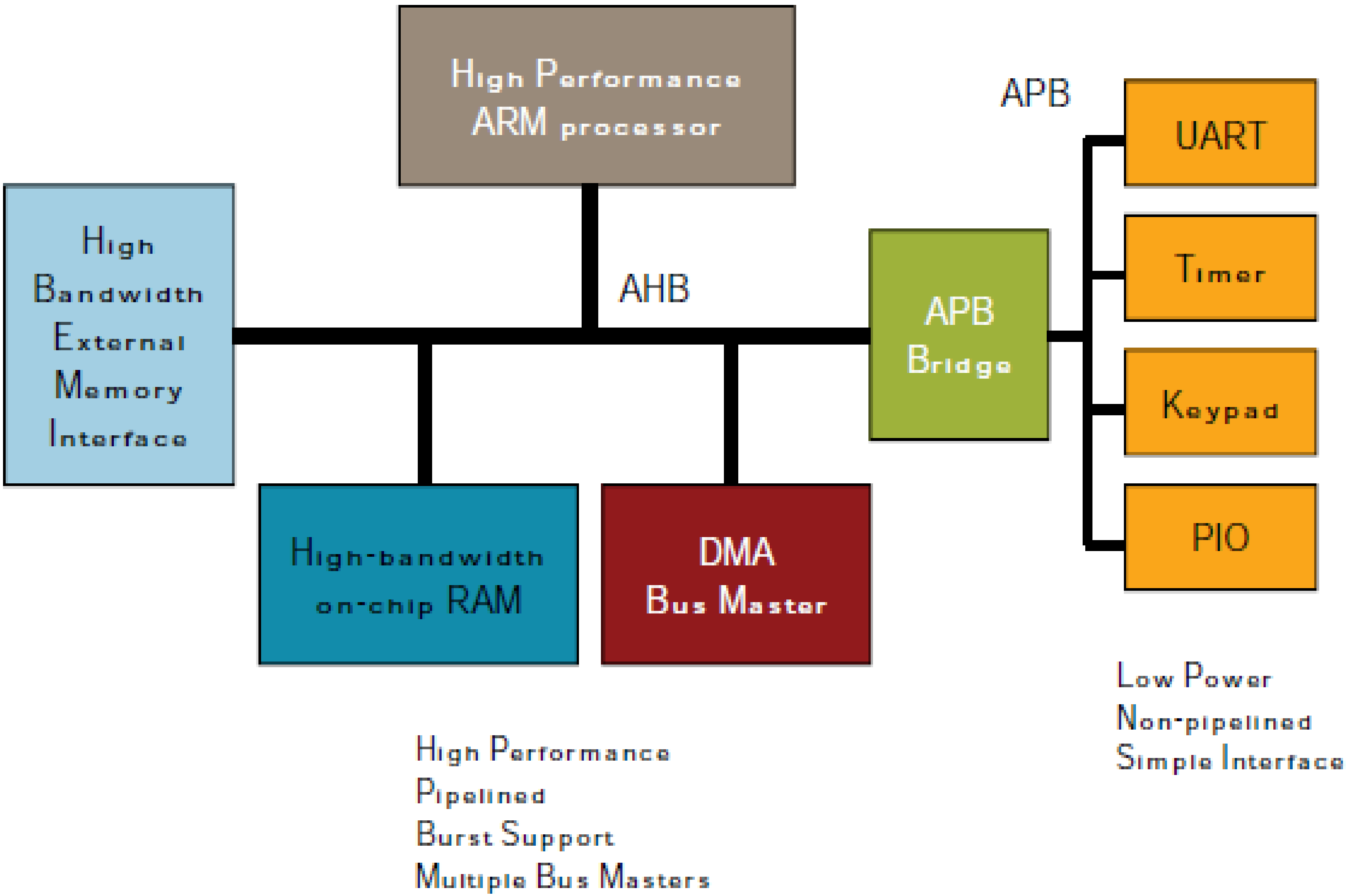
Advanced High-Performance Bus AHB



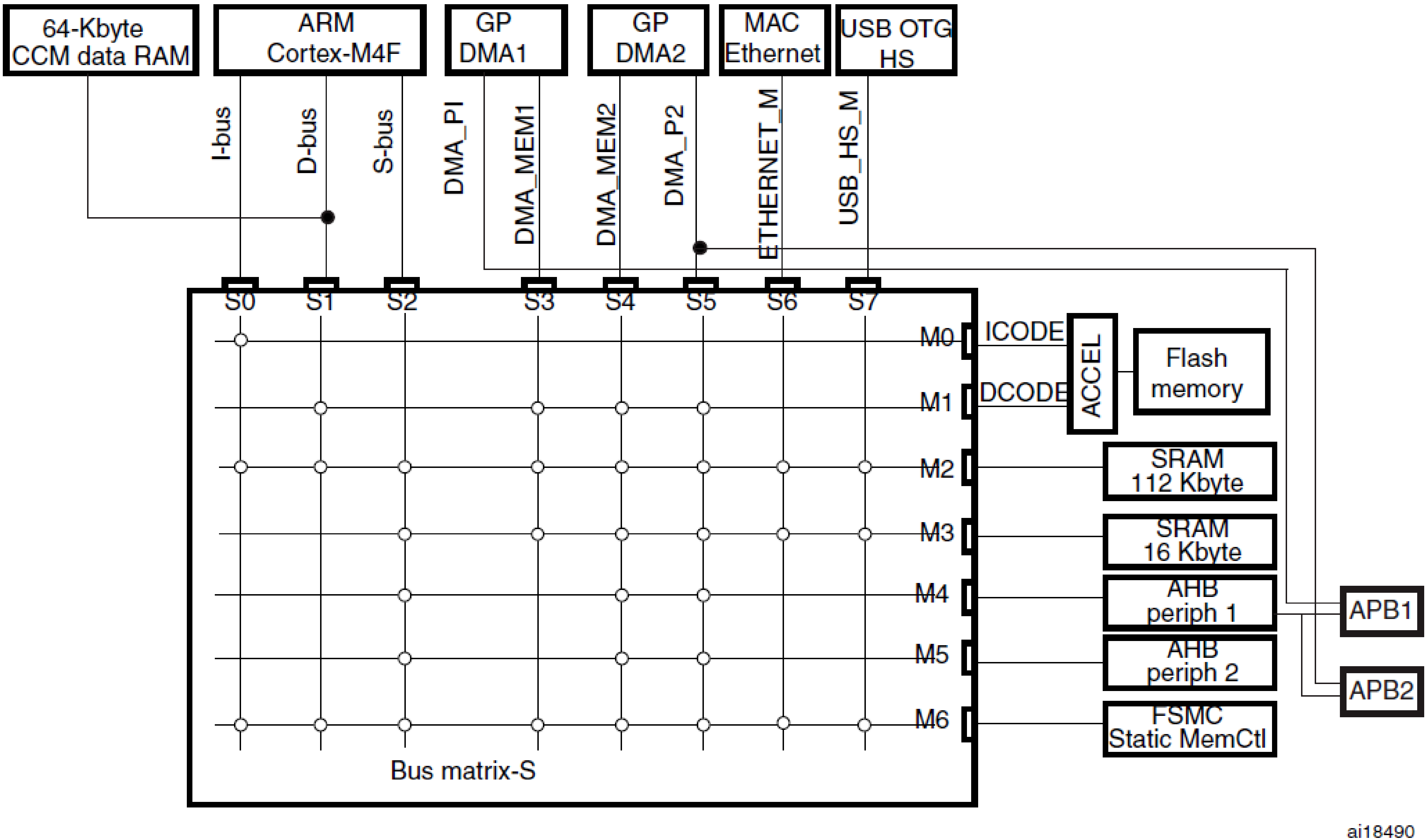
NXP LPC800 CM0+



ARM Cortex M4 32-bit Controller



ARM Cortex M4 Bus Systems



STM32F7x5 and STM32F7x7 Multicore MPU